STAFF REVIEW · H446 140 source models

Separate from the eight-page pupil workbook. This copy is in a public repository and is NOT access controlled. Check technical correctness before use.

Printed question mapping: Booklet Q1 = original online Q1; Booklet Q2 = original online Q2; Booklet Q3 = original online Q3

Do Now — source answers

1. Every row in the target table can be updated.

2. Tracking variable state changes as an algorithm runs.

3. Eight.

4. Stack.

I Do — source worked steps

1. Customers 7 and 9 — Identify the relevant information and make a prediction.

2. Order requests ID 12 — Explain the change using correct technical terms.

3. Reject orphan reference — The DBMS rejects the foreign key value 12, unless a matching Customer row is created or the reference is changed. Policies for deleting parent rows may RESTRICT, CASCADE or SET NULL, where permitted and configured.

We Do — guided CHECK

1. Frequent ISBN lookup
Identify the relevant information and make a prediction.
Indexed search path
Explain the change using correct technical terms.
Write and space overhead
An index on ISBN can support quicker exact-match retrieval, especially for large data, but occupies space and requires maintenance after insert/update/delete. A UNIQUE constraint is appropriate if ISBN must be unique in this table.

You Do — source mark schemes

1. Loan FK references existing Borrower key [2], prevents dangling child rows [2], example insertion/deletion rule [2].

2. Faster key lookups [2], faster ordered/range operations where supported [2], storage or index-update overhead [2].

3. RESTRICT denies parent deletion [2]; CASCADE removes dependent loans [2]; loss/history risks [2]; reasoned chosen retention/archival policy [2].

Exit Ticket — source answer

1. False. Uniqueness requires a unique index or constraint.